# PROBABILITY & STATISTICS



Priority: state support → normalize → choose sums / integrals → condition correctly → check total probability and endpoints.

## 01 Data summaries

$$\bar{x} = \frac{\sum_i x_i}{n}, \quad s^2 = \frac{\sum_i x_i^2 - n\bar{x}^2}{n-1}, \quad s = \sqrt{s^2}$$

**Sort first.** Median = middle value (average the middle two if n is even). IQR = $Q_3 - Q_1$. Median/IQR resist outliers; mean/SD do not.

$$Y = a + bX: \quad E[Y] = a + bE[X], \quad \mathrm{SD}(Y) = |b|\mathrm{SD}(X)$$

For nonlinear g, usually E[g(X)] ≠ g(E[X]).

Adding a constant leaves spread unchanged. Normal shape: 68%, 95%, 99.7% within 1, 2, 3 SD.

## 02 Events: translate before calculating

$$P(A^c) = 1 - P(A), \quad P(A \setminus B) = P(A) - P(A \cap B)$$
$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

**At least one** = 1 - none. If A is contained in B:

$$A \cap B = A, \quad A \cup B = B, \quad P(B \cup A^c) = 1$$

## 03 Counting: order? replacement?

$$\text{Ordered: } n^k \text{ (replace)}; \quad (n)_k = \frac{n!}{(n-k)!} \text{ (no)}$$
$$\text{Unordered: } \binom{n+k-1}{k} \text{ (replace)}; \quad \binom{n}{k} \text{ (no)}$$
$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}; \quad \text{repeated objects: } \frac{n!}{\prod_i n_i!}$$

Multiply choices; add disjoint cases. **Equally likely only:** P(A) = favorable / total. Fair coins: $2^n$ sequences, C(n,k) with k heads.

## 04 Inclusion-exclusion and collisions

$$P(A \cup B \cup C) = \sum P(A_i) - \sum_{i<j} P(A_i \cap A_j) + P(A \cap B \cap C)$$

Add singles, subtract pairs, add triples, ... . For n independent uniform choices among d labels:

$$P(\text{collision}) = 1 - \frac{(d)_n}{d^n} \quad (n \le d)$$

$$P(\text{all } d \text{ seen}) = \sum_{j=0}^{d} (-1)^j \binom{d}{j} \left(1 - \frac{j}{d}\right)^n$$

## 05 Conditional probability and Bayes

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A \mid B)P(B)$$

P(B) > 0. Given B, restrict to B. Without replacement: update numerator and denominator after each draw.

$$P(E) = \sum_i \pi_i L_i, \quad P(H_i \mid E) = \frac{\pi_i L_i}{\sum_j \pi_j L_j}$$

$H_i$ partition outcomes. Prior $\pi_i = P(H_i)$; likelihood $L_i = P(E \mid H_i)$; posterior = updated probability.

**Repeated tests / same sent bit [both papers]**

π = P(H), a = P(+ | H), b = P(+ | $H^c$). For m tests independent **given the same H**:

$$P(m+) = \pi a^m + (1-\pi) b^m$$
$$P(H \mid m+) = \frac{\pi a^m}{\pi a^m + (1-\pi) b^m}$$

**Mix after multiplying.** Do not square P(+). Sequential update: posterior becomes next prior.

## 06 PMF, PDF and CDF [both papers]

**Support first.** PMF: p ≥ 0, Σp = 1. PDF: f ≥ 0, integral = 1. Density height may exceed 1.

$$F(x) = P(X \le x) = \sum_{u \le x} p(u) \quad \text{or} \quad \int_{-\infty}^{x} f(u)\,du$$

CDF: nondecreasing, right-continuous, limits 0 and 1. Below support F = 0; above support F = 1.

**Endpoints matter when there are atoms**

$$F(a^-) = P(X < a), \quad P(X = a) = F(a) - F(a^-)$$
$$P(a < X \le b) = F(b) - F(a)$$
$$P(a \le X \le b) = F(b) - F(a^-)$$
$$P(a < X < b) = F(b^-) - F(a)$$
$$P(a \le X < b) = F(b^-) - F(a^-)$$
$$P(X > a) = 1 - F(a), \quad P(X \ge a) = 1 - F(a^-)$$

Integers: p(k) = F(k) - F(k-1); P(X ≥ k) = 1 - F(k-1). CDF is flat between jumps.

**Continuous:** P(X = a) = 0; endpoints do not matter. f = F′ where differentiable. For mixed laws, the derivative misses atoms.

**Piecewise CDF recipe**

Split at formula changes. Add **all prior area** to each new piece. Check joins and tails.

$$F(x) = F(c) + \int_c^x f(u)\,du \quad \text{within the next piece}$$

Quantile $q_p$ = inf{x : F(x) ≥ p}; if continuous and strictly increasing, solve $F(q_p) = p$. Median: p = 1/2.

## 07 Joint tables, marginals, conditioning

$$\sum_x \sum_y p(x,y) = 1, \quad p(x,y) \ge 0$$
$$p_X(x) = \sum_y p(x,y), \quad p_Y(y) = \sum_x p(x,y)$$
$$p_{X \mid Y}(x \mid y) = \frac{p(x,y)}{p_Y(y)} \quad [p_Y(y) > 0]$$

**Tables (X rows, Y columns):** missing cell = margin - others. X = Y: sum diagonal. Given Y = y: column / column total; sums to 1.

$$P((X,Y) \in D) = \sum_{(x,y) \in D} p(x,y)$$

## 05 / 07 Independence checks

$$A \perp B \iff P(A \cap B) = P(A)P(B)$$

Pairwise: check every pair. **Mutual:** product rule for every subset, including triples. Disjoint positive-probability events are dependent.

$$X \perp Y \iff p(x,y) = p_X(x)p_Y(y) \quad \text{for all } (x,y)$$

Discrete: one mismatch suffices. Conditional ≠ unconditional independence.

**Conditional category counts**

$$X \mid (Y = k) \sim \mathrm{Bin}\left(n-k, \frac{p_X}{1-p_Y}\right)$$

n iid categorical trials; X,Y count distinct categories. Given k Y outcomes, renormalize remaining probabilities.

## 08 Joint densities and integration

$$\iint_D f(x,y)\,dA = 1, \quad P((X,Y) \in R) = \iint_{R \cap D} f\,dA$$

For f = cg: c = 1/(integral of g over D). For cx + 1, integrate the **whole expression** and solve.

$$f_X(x) = \int_{\text{allowed } y} f(x,y)\,dy, \quad f_Y(y) = \int_{\text{allowed } x} f(x,y)\,dx$$
$$f_{X \mid Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)} \quad [f_Y(y) > 0]$$

**Keep the requested variable; integrate out the other.** State support; zero elsewhere. Conditional density integrates to 1.

**Sketch D, then choose slices [both papers]**

Inner variable moves, outer is fixed. Reverse bounds for the **same region**.

$$x, y \ge 0,\ x + y \le 1: \quad \int_0^1 \int_0^{1-x} f\,dy\,dx = \int_0^1 \int_0^{1-y} f\,dx\,dy$$
$$0 \le y \le x \le 1: \quad \int_0^1 \int_0^{x} f\,dy\,dx = \int_0^1 \int_y^{1} f\,dx\,dy$$

If f depends only on x: inner y integral = f(x) × y-width. Uniform on D: c = 1/area(D), P(R) = area(R ∩ D)/area(D).

$$X \perp Y \iff f(x,y) = f_X(x) f_Y(y) \quad \text{a. e.}$$

Factorization includes support; a positive triangular density fails. **For jointly normal X,Y:**

$$X_i \sim N(\mu_i, \sigma_i^2), \qquad \rho = 0 \iff X \perp Y$$
$$Y \mid X = x \sim N\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1-\rho^2)\right)$$

N(mean, variance). Zero correlation alone does not imply independence in general.

## 09 Expectations, moments and variance

$$E[g(X)] = \sum_x g(x)p(x) \quad \text{or} \quad \int g(x)f(x)\,dx$$
$$E[g(X,Y)] = \sum_{x,y} g(x,y)p(x,y) \quad \text{or} \quad \iint gf\,dA$$
$$\mu = E[X], \quad \mathrm{Var}(X) = E[X^2] - \mu^2, \quad \sigma = \sqrt{\mathrm{Var}(X)}$$
$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

Linearity needs no independence. If symmetric about m, E[X] = m (if finite).

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$$
$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y], \quad \rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$
$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$$

Independent: E[XY] = E[X]E[Y], covariance = 0, variances add. $E[X^2]$: square **values**, not probabilities.

$$\text{skew} = \frac{E[(X-\mu)^3]}{\sigma^3}, \quad \text{excess kurtosis} = \frac{E[(X-\mu)^4]}{\sigma^4} - 3$$

## Calculus used in these papers

$$\int x^r dx = \frac{x^{r+1}}{r+1} + C\ (r \ne -1), \quad \int \frac{dx}{x} = \ln|x| + C$$
$$\int e^{ax} dx = \frac{e^{ax}}{a} + C\ (a \ne 0), \quad \int u\,dv = uv - \int v\,du$$

Substitute u = g(x), du = g′(x)dx; change limits too. At ∞, take a limit. $\int_1^\infty x^{-r} dx$ converges iff r > 1; then equals 1/(r-1).

## 10 MGF and CGF [2025 Q2, Q5]

$$M_X(t) = E[e^{tX}] = \sum_x p(x)e^{tx} \quad \text{or} \quad \int e^{tx} f(x)\,dx$$
$$M(0) = 1, \quad M^{(k)}(0) = E[X^k]$$
$$\mu = M'(0), \quad \sigma^2 = M''(0) - [M'(0)]^2$$
$$M_{aX+b}(t) = e^{bt} M_X(at), \quad M_{X+Y}(t) = M_X(t)M_Y(t) \text{ (indep.)}$$

Must be finite on an **open interval around 0**. Finite support always works. MGFs near 0 identify the law. Discrete $e^{tx}$ coefficient = P(X=x).

**Heavy-tail test:** f = $2/x^3$ on x ≥ 1 has $E[X^2] = 2\int_1^\infty dx/x = \infty$, so no MGF near 0. M(0) = 1 alone is insufficient.

$$K(t) = \ln M(t), \quad K'(0) = \mu, \quad K''(0) = \sigma^2$$

## 10 Recognize the model; q = 1 - p

E = mean; V = variance. For geometric/NB: 0 < p ≤ 1. Trial counts and population sizes are integers.

**Bernoulli(p): one success indicator**

$$p_X(k) = p^k q^{1-k}, \ k = 0, 1; \quad E = p, \quad V = pq$$

**Binomial(n,p): successes in n iid trials**

$$p_X(k) = \binom{n}{k} p^k q^{n-k}, \ 0 \le k \le n; \quad E = np, \ V = npq$$
$$M(t) = (q + pe^t)^n$$

**Geometric(p): trials through first success**

$$p_X(k) = pq^{k-1}, \ k \ge 1; \quad E = 1/p, \quad V = q/p^2$$
$$P(X > k) = q^k, \quad M(t) = \frac{pe^t}{1 - qe^t}$$

**Negative binomial(r,p): trials through rth success**

$$p_X(k) = \binom{k-1}{r-1} p^r q^{k-r}, \ k \ge r; \quad E = r/p, \ V = rq/p^2$$

Count trials **including final success**. MGFs need $qe^t < 1$; NB MGF = [geometric MGF]$^r$.

**Poisson(λ): count in an interval, λ = rate × length**

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \ k \ge 0; \quad E = V = \lambda$$
$$M(t) = e^{\lambda(e^t - 1)}; \quad \mathrm{Bin}(n,p) \approx \mathrm{Pois}(np) \ (n \text{ large}, p \text{ small})$$

**Hypergeometric(N,K,n): n draws, no replacement**

$$p_X(k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \quad E = n\frac{K}{N}$$
$$V = n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1} \quad (N > 1)$$

K successes in N objects; max(0,n-N+K) ≤ k ≤ min(n,K). Binomial approximation only if n/N is small.

**Uniform: equal weights / constant density**

Integers a,...,b: m=b-a+1, P=1/m, E=(a+b)/2, V=(m²-1)/12. Continuous U(a,b): f=1/(b-a), same E, V=(b-a)²/12.

**Multinomial: n iid trials, category probabilities $p_i$**

$$P(\mathbf{X} = \mathbf{k}) = \frac{n!}{\prod_i k_i!} \prod_i p_i^{k_i}, \quad \sum_i k_i = n, \ \sum_i p_i = 1$$

Each $X_i \sim \mathrm{Bin}(n, p_i)$; combine categories by adding their probabilities. All PMF counts k are integers; impossible values have probability 0.